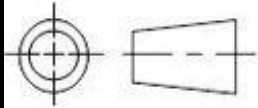


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Revision History

Rev.	Description	Date	Approvals
B	Updating the drawing template to comply with GE standards	2025-05-28	Created By: Saibabu Lakkamraju Approved By: Evilene James Ortega

THIRD ANGLE PROJECTION



THIS DRAWING TO BE REVISED IN MICROSOFT WORD ONLY. THE SOURCE FILE IS STORED IN THE EDMS

CYBER SECURITY

LIST OF COMPLEMENTARY DOCUMENTS

ITEM	DESCRIPTION	DWG. NO.

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Similar To: N/A	Created By: Saibabu Lakkamraju	MLI N/A
First Made For: N/A	Approved By: Evilene James Ortega	IEC61355 No: N/A
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1.0 INTRODUCTION

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This document describes the proposed Customer witness test (CWT) of the Cyber Security Management System.

The Customer witness test will take place in GE Vernova Facilities, during a period of **<NUMBER OF DAYS>** days. Testing will include hardware, software, and a documentation review of the system. The intent of the test is to demonstrate that the system under test operates as designed. This document will be used as the approval document to provide a record of the test activities.

At the end of the CWT a summary of the test results and corrective actions will be added to this document and archived. Any items that cannot be resolved prior to shipment will be addressed by engineering and communicated to the site installation team prior to the unit outage by the most effective means possible.

1.1 Punch List Tabulation of Problem Reports

During the CWT test personnel will document any problems that require corrective action. The tester will identify their name, the date that the problem was found, any document references, and a description of the problem.

When problems are discovered, they will be reviewed with the engineer for that area. Problems and non-conformances shall be formally logged in Section AGREED RESIDUAL ACTIONS.

The CWT owner or an engineer/technician under the direction of the CWT owner will implement all corrections. Problems will be considered resolved only when the engineer responsible for that area of the test signs them off. All problem reports must be resolved prior to authorization for shipment.

1.2 Error Correction / Retest

The timing of error corrections and retests will depend on the type and scope of the problems encountered. Problems that can be easily corrected will be corrected when identified and recorded in Section AGREED RESIDUAL ACTIONS. Problems that require more time to correct may be resolved and/or re-tested prior to shipment or as agreed between the parties. Residual Actions may be moved to a Project Open Items List as a method of resolution as to not hold up completion of testing or shipment.

Replacement of any hardware, firmware or changes to the software configurations may require that the affected components or code be retested to ensure that the design remains valid as determined by the Test Engineer. The retest will be performed using the relevant sections of this CWT procedure.

1.3 Change Management

A change management log will be used for the duration of the CWT. His log will track any changes made to the system that vary from the original configuration. All changes need to be approved by GEV and the customer. Logs will be recorded in the "Comments and Observations" of each task.

1.4 Reference Documents

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The following is a list of reference documents used in conjunction with the Customer witness test the latest revisions of these documents will be used at the time of the test. The GE Cyber Engineer shall be responsible for obtaining the latest revisions.

- Network Topology (<DOCUMENT NUMBER>).
- GEV Controls Ports and Services Document (132M6452).
- Pre-FAT Ports and Services Scan Results.

Note: This document is a complete checklist for security features. A project CWT checklist will be generated for the features included with a specific project."

1.5 OTArmor Machine Roles

1.5.1 DC1, DC2:

Primary and Secondary Domain Controllers. These machines manage authentication and configurations for other assets.

1.5.2 DR1:

Disaster Recovery. This machine hosts the CatTools and Acronis applications, which are used for backups and recovery.

1.5.3 PM1:

Patch Management. This machine hosts the Ivanti application, which is used for deploying Windows patches.

1.5.4 SP1:

System Protection. This machine hosts the Trellix Electronic Policy Orchestrator (ePO). The ePO is an agent-based software that manages Endpoint Security (Antivirus), SolidCore (Application Whitelisting) and DLP (Data Loss Prevention).

1.5.5 SIEM:

Security Information and Event Management. A Trellix non-Windows machine that is accessed via a web browser. OTArmor assets are configured to forward event logs to be centrally viewed and managed.

1.5.6 LM1:

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A log management system is a system used to collect, store, process, and analyze log data from various IT systems and applications

1.5.7 NM1:

Centralized platform used to monitor, manage, and optimize a network's performance, security, and resource utilization

2.0 **REQUISITION INFORMATION**

Customer: SAUDI ELECTRICITY COMPANY CENTRAL REGION
End User: SEC
Site Name: RIYADH PP11
Project number #:F6900G6

3.0 **EQUIPMENT INSPECTION**

The following sections provide a brief description of the inspection / test. It also permits the recording of comments and allows sign off on each test.

3.1 **Equipment Inspection**

This task performs a visual inspection system or HMIs.

Description	Comments & Observations
Perform a visual inspection of the system or HMIs.	
Expected Result to pass:	
All components are present and represented in the system drawings.	

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3.2 Remote Desktop to HMIs (Virtual machines only)

This task defines if the HMIs will need to have Remote Access enabled. If the HMI's to be installed at site are not present during the CWT, this task will be performed at site.

If the HMIs are present during the CWT and Remote desktop access is requested, in each HMI:

- 3.2.1 Navigate to "Control Panel -> System -> Remote Settings".
- 3.2.2 Select "Allow remote connections to this computer".
- 3.2.3 If possible, select "Allow connections only from computers running Remote Desktop with Network Level Authentication (recommended)". There may be cases when connecting to an older computer may need to leave this unchecked.
- 3.2.4 The following services must be turned on for remote desktops to work:
 - 3.2.4.1 Navigate to "Control Panel -> Administrative Tools -> Services".
 - 3.2.4.2 Find the "Remote Desktop Services Service", enable it and place it in automatic.
 - 3.2.4.3 Find the "Remote Desktop Services User Mode Port Redirector", enable it and place it in automatic. Start this service.
- 3.2.5 Verify Remote Desktop is added in the Windows Firewall:
 - 3.2.5.1 Navigate to "Control Panel -> Windows Firewall -> Allow an app or feature through Windows Firewall".
 - 3.2.5.2 On the list displayed look for "Remote Desktop" and make sure "Domain" and "Private" are checked. If not, proceed to click on "Change Settings" and check "Domain" and "Private" and click "OK".
- 3.2.6 Verify that RDP connections are successful.

NOTE. If RDP is not required by the customer or not needed anymore, disable Remote Desktop in the Windows Firewall.

Description	Comments & Observations
If required, verify Remote Desktop is enabled on all HMIs.	
Expected Result to pass:	
Verify we can Remote Desktop into the HMI's configured.	

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4.0 RADIUS

4.1 RADIUS Functionality

This task is to verify the network switches that have been entered in Radius, this test will be performed Domain all Controllers:

- 4.1.1 Log into one of the DC using an administrator account and Launch the Server manager, open "Tools->Network Policy Server".
- 4.1.2 Expand "RADIUS Clients and Servers" and select "RADIUS Clients" and verify the Network Switch are present as per Network Topology.
- 4.1.3 Disable the Network Policy Server on Domains.
- 4.1.4 Attempt to authenticate to the Network Switch, verify that the radius services can NOT successfully authenticate the "GESwitch & GESwitchOper" account.
- 4.1.5 Attempt to log onto the Network Switch using the local account 'Admin' and verify it is successful.
- 4.1.6 Enable the Network Policy Server on Domains.
- 4.1.7 Attempt to authenticate to the Network Switch, verify that the radius services can successfully authenticate the "GESwitch & GESwitchOper" account.
- 4.1.8 Attempt to log onto the Network Switch using the local account 'Admin' and verify it is unsuccessful

Description	Comments & Observations
Review RADIUS in Active Directory on all Domain Servers to verify Network Switch is entered. When "Network Policy Server" is Disable verify: • Unsuccessful access occurs with "GESwitch & GESwitchOper" when trying to login into Network Switch. • Successful access occurs with Local "Admin" Account when trying to login into Network Switch. When "Network Policy Server" is started, verify: • Unsuccessful access occurs with Local "Admin" when trying to login into Network Switch. • Successful access occurs with "GESwitch & GESwitchOper" Account when trying to login into Network Switch.	
Expected Result to pass:	

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- Confirm Domain Servers have the Network Switch(es) configured in Radius.
- Confirm you can authenticate with the GESwitch & GESwitchOper user account in the Network Switch when the Network Policy Server is running. Confirm authentication fails when the Network Policy Server is NOT running.
- Confirm you can authenticate with the Local Admin user account in the Network Switch when the Network Policy Server is NOT running. Confirm authentication fails when the Network Policy Server is running.

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4.2 Verify RADIUS Account Login

This task is to verify the Network Switch can be accessed with a Network Domain Account 'GESwitch & GESwitchOper':

- 4.2.1 Log into the DR1 using an administrator account and launch the PuTTY shortcut on the desktop.
- 4.2.2 Enter the information to connect to a Network Switch and login with the "GESwitch & GESwitchOper" credentials.

Description	Comments & Observations
Verify the Network Switch can be accessed with the Active Directory account 'GESwitch & GESwitchOper'.	
Expected Result to pass:	
Successful login with GESwitch & GESwitchOper account.	

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4.3 Verify RADIUS Account Lockout

This task is to verify a domain account will be locked after 10 failed attempts using RADIUS:

- 4.3.1 Log into the DR1 using an administrator account and launch the putty shortcut on the desktop, Provide IP address/Hostname details and click on ok to open session.
- 4.3.2 Enter the correct Username but an incorrect password for the GESwitch & GESwitchOper account. Make sure you try these 10 times. (If Putty shows an error, please close Putty and open again).
- 4.3.3 After 10 tries, type the correct password. It should still report you are not able to log into the system.
- 4.3.4 Log into any of the DC using an administrator account. Open Server Manager, on the top menu chose "Tools-> Active Directory Users and Computers".
- 4.3.5 A new window will appear (Active Directory Users and computers), expand "Domain Name(HMI.local)->GEV Controls->GEV Users and Computers". Look for GESwitch & GESwitchOper Account and double click it.
- 4.3.6 Click on the "Account" tab, you will see the account is being locked. Check "Unlock Account" and click OK.
- 4.3.7 Try to login into Network Switch using Putty with the correct Credentials and password. Confirm the account can login successfully.

Description	Comments & Observations
Verify the GESwitch & GESwitchOper account is locked when entering 10 bad passwords. Reset the account, log in successfully.	
Expected Result to pass:	
Verify Account locks after 10 fail attempts. Also verify you can login successfully after resetting the account.	

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5.0 Verify NIDS Configuration (Optional)

- 5.1.1 Generate several NIDS strings in the system (Test-NIDS2.ps1) Open file explorer and type this path: [\\CA1\Admin Procedures\OTArmor](#) on any of the HMIs and take note of the time they were executed. This PowerShell will simulate an attack in the system.
- 5.1.2 Log into the firewall and under Network->Interfaces check the port is configured for NIDS.
- 5.1.3 Also, under Log&Report->Intrusion Prevention, verify you got the logs for the attack (Test-NIDS2.ps1).
- 5.1.4 "GEV NIDS TEST STRING" is the string that will be listed in the logs. Verify time matches with the ones you executed and from the devices they were executed.

Note. This step will be verified a 2nd time if you have LM1.

Description	Comments & Observations
Verify "GE NIDS TEST STRING" can be found in the local logs after executing Test-NIDS2.ps1.	
Expected Result to pass:	
Verify the Firewall is receiving "BHGE NIDS TEST STRING" logs and they match the ones that were executed with Test-NIDS2.ps1.	

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6.0 Cat Tools

6.1 **Verify Network Switch is configured in CatTools**

This task is to verify Network Switch configured in CatTools:

- 6.1.1 Log into the DR1 using an administrator account and launch the CatTools shortcut on the desktop.
- 6.1.2 On the left panel you, under the “Devices” tab, a list of devices will be listed. Please verify the Network Switch is listed with its correct IP Address.

Description	Comments & Observations
Verify the network switch is listed in CatTools.	
Expected Result to pass:	
In the list of devices, the network switch(es) will be present with correct IP Address(es).	

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6.2 Verify Network Switch Backup Task in CatTools

This task is to verify network switch can be backed up using CatTools:

- 6.2.1 Log into the DR1 using an administrator account and launch the CatTools shortcut on the desktop.
- 6.2.2 On the right Panel, under “Activities” Tab, look for the “Backup Activity”.
- 6.2.3 Right click on the task and choose “Run Now”.
- 6.2.4 On the bottom panel, you will see and “Activity Log”, it will inform you how the activity is running and if it founds any error while executing. (Please be patient, getting a backup can take up to 5 to 10 minutes). Once the task is finished, the logs will report if the task is finished successfully or with errors.
- 6.2.5 When task finishes running successfully, go to the “File Explorer” and browse to the following path: “C:\Backups\Switches\Config”. It should list a file with the name “Config.Current.Running. [Name of Network Switch].txt”.

Description	Comments & Observations
Verify the Network Switch(es) can be backed up in CatTools.	
Expected Result to pass:	
Backup tasks will run successfully, and a file will be created under C:\Backups\Switches.	

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6.3 Verify Configuration Changes are Identified

This task is to verify Network Switch changes can be tracked using CatTools:

- 6.3.1 Log into DR1 using an administrator account.
- 6.3.2 Execute a backup task on the Network Switch.
- 6.3.3 After the backup task is complete, log into the Network Switch and perform a configuration change.
- 6.3.4 Re-execute the backup task in CatTools.
- 6.3.5 Open File Explorer and navigate to “C:\Backups\Switches\Reports\ConfigChanges”.
- 6.3.6 A list of files will be shown, open the latest one by double clicking it.
- 6.3.7 Review the Report and verify that the report identifies what was changed on the Network Switch.

Description	Comments & Observations
Use the reports function to verify any network switch changes that can be documented.	
Expected Result to pass:	
The CatTools application will identify the changes made to the device under test.	

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6.4 Verify the Network Switches can be restored using CatTools

This task is to verify the Network Switches can be restored using CatTools:

- 6.4.1 Log into the DR1 using an administrator and launch the CatTools shortcut on the desktop.
- 6.4.2 Verify the selected Config file from the “C:\Backup\Switches\Config” folder is put into the “C:\Backup\Switches\Restore” folder
- 6.4.3 On the right panel in CatTools, under the “Activities” tab, a list of Activities will be listed. Double click on the “Restore Network Switch”.
- 6.4.4 Verify the Network Switch gets restored to the selected configuration.

Description	Comments & Observations
Verify the Network Switches can be restored to a previous configuration using CatTools.	
Expected Result to pass:	
Verify the Network Switch has been restored to its selected configuration successfully.	

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6.5 Verify Network Switches have a customer approved banner during sign in (Optional)

This task is to verify the Network Switches have a customer approved banner during sign in:

6.5.1 Log into the DR1 using an administrator.

6.5.2 Open File Explorer and navigate to "C:\Backup\Switches\Config". Open the config of the Network Switches and review the file for the customer approved banner.

6.5.3 Verify the Network Switch has the customer approved banner.

Description	Comments & Observations
Verify the Network Switch displays the customer approved banner during sign in.	
Expected Result to pass:	
Verify the Network Switch displays the customer approved banner.	

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6.6 Verify the Network Switches are synced to NTP source

This task is to verify the Network Switches sync to the proper NTP source:

- 6.6.1 Log into the DR1 using an administrator account and launch the CatTools shortcut on the desktop.
- 6.6.2 On the left panel, under the “Devices” tab, a list of devices will be listed. Double click on the Network Switch device listed.
- 6.6.3 Login to the Network Switch and verify the switch has the correct NTP Source at the end of the config.
 - 6.6.3.1 Verify NTP status of the switch “**show ntp status**”
 - 6.6.3.2 Verify NTP associations of the switch “**show ntp associations**”
 - 6.6.3.3 Verify Switch time “**show Clock**”

Description	Comments & Observations
Verify the Network Switches are synced to the correct NTP Source at the end of the config.sh	NTP Source:
Expected Result to pass:	
Verify the Network Switches have the correct NTP source added successfully at the end of the config.	

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7.0 NTP

7.1 Verify NTP Synchronization

This task is to verify if the domains are synced to the proper NTP Server:

7.1.1 Log into the DC using an administrator account.

7.1.2 Open the Admin Command Prompt and type: "w32tm /query /peers" <Enter>

7.1.3 Verify the system is pointing to the correct NTP IP Address.

Description	Comments & Observations
Verify DC1/ DC2 is synced to the proper time source.	NTP Source: _____
Expected Result to pass:	
Verify the Domain Server is pointing at the correct time source.	

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7.2 Verify Servers Sync to NTP Server (HMI's Excluded)

This task is to verify if the Servers are synced to the Active Domain Controller:

7.2.1 Log into each Servers using an administrator account.

7.2.2 Open the Admin Command Prompt and type: "w32tm /query /peers" <Enter>.

7.2.3 Verify the system is pointing to the Active Domain Controller IP Address.

Description	Comments & Observations
Verify all Servers are synced to DC1.	
Expected Result to pass:	
Verify the servers are pointing at the correct Active Domain Controllers.	

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8.0 Account Management

8.1 Default/Domain Accounts

This task is to verify only the applicable Users are enabled in the Domain and in HMI/Servers/NTP/Default Switches:

- 8.1.1 Log into any of the DC using an administrator account. Open Server Manager, on the top menu chose "Tools-> Active Directory Users and Computers".
- 8.1.2 A new window will appear (Active Directory Users and computers), expand "Domain Name (HMI.local)->GEV Controls->GEV Users and Computers"Select "Tools->Active Directory Users and Computers" and review the Windows user accounts present.
- 8.1.3 Verify only necessary accounts are present and enabled. Default windows accounts not required are disabled or removed.
- 8.1.4 To check the local Users on the HMIs/Servers:
 - 8.1.4.1 Log into the HMI and open "Control Panel -> All Control Panel Items-> Administrative Tools -> Computer Management -> Local Users and Groups -> Users".
 - 8.1.4.2 Review the windows user accounts present.
 - 8.1.4.3 Verify only necessary accounts are present and enabled. Default windows accounts not required are disabled or removed.

Description	Comments & Observations
Verify Windows default accounts have been deleted, disabled or renamed on the HMIs, Default Switches, NTP Server and Servers.	
Expected Result to pass:	
All the Accounts that are not going to be used should be disabled. In the comments list all the accounts disabled.	

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8.2 Verify Operator Account

This task is to verify the Operator account settings to comply with Customer needs:

8.2.1 Log into the HMI using the GEOper account.

8.2.2 Verify the privileges that the account has:

8.2.2.1 Verify there is no access to the Command Prompt, no ability to Navigate to the hard drives, and cannot Open the Control Panel.

8.2.3 If needed, make changes on the account and describe them under “Comments & Observations”.

Description	Comments & Observations
Verify ‘Operator’ account has least amount of user privileges. Accounts should not have any Admin rights.	
Expected Result to pass:	
Verify Operator Account does not have an Administrator Account.	

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8.3 Verify Engineer/Maintenance Account

This task is to verify the Engineer/Maintenance account settings comply with Customer needs:

8.3.1 Log into the HMI using the GEMaint account.

8.3.2 Verify the privileges of the account by:

8.3.2.1 Opening the Command Prompt, navigate to the C: drive, Verify the. TCW is not in "Read-Only" mode, Open Control Panel and verify user has access and can make changes.

8.3.3 If needed, make changes on the account and describe them under "Comments & Observations".

Description	Comments & Observations
Verify 'Engineer/Maintenance' account has the right user privileges. Account should have Maintenance rights.	
Expected Result to pass:	
Verify Engineer/Maintenance Account has Command Prompt, Control Panel Access, can make changes in the E: Drive and. TCW.	

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8.4 Verify Shared Accounts

This task is to verify all shared accounts:

- 8.4.1 Log into any of the DC using an administrator account. Open Server Manager, on the top menu chose "Tools-> Active Directory Users and Computers".
- 8.4.2 A new window will appear (Active Directory Users and computers), expand "Domain Name (HMI.local)->GEV Controls->GEV Users and Computers" Select "Tools->Active Directory Users and Computers" and identify which users are shared Accounts. (A shared Account is when more than one person can access and login with an account).

Standard default shared accounts are mentioned below

- GEAdmin
- GEMaint
- GEOper

Description	Comments & Observations
Identify any shared accounts and if they are needed.	
Expected Result to pass:	
List under comments all the shared Accounts found.	

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8.5 Service Accounts

This task is to verify all service accounts:

8.5.1 Log into any of the DC using admin account and launch “Server Manager”.

8.5.2 Select “Tools -> Group Policy Management”.

8.5.2.1 Under “Domain Name (HMI.local)-> GEV Controls->Domain Controllers ->GEV DCx Hardening Policy”. Select “Settings” tab and review the service accounts listed under “Deny Log on Locally”. Verify the User Group “Deny Log Locally” is listed.

8.5.2.2 Under “Domain Name (HMI.local)-> GEV Controls -> HMI->Windows 10 HMI -> GEV Win10 Hardening Policy”. Select “Settings” tab and review the service accounts listed under “Deny Log on Locally”. Verify the User Group “Deny Log Locally” is listed.

8.5.2.3 Under “Domain Name (HMI.local)-> GEV Controls -> OTArmor Servers-> GEV Server OS Hardening Policy”. Select “Settings” tab and review the service accounts listed under “Deny Log on Locally”. Verify the User Group “Deny Log Locally” is listed.

8.5.3 Select “Tools -> Active Directory Users and Computers”. Domain Name (HMI.local)->GEV Controls->GEV Users and Computers” Select “Tools->Active Directory Users and Computers”. Double click “Deny Log Locally” User group and choose “Member of” tab. Verify the list of users.

Description	Comments & Observations
Identify any accounts associated with automated functions.	
Expected Result to pass:	
List under comments all the service Accounts found.	

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8.6 **Verify Service Account Restrictions**

- 8.6.1 This task is to verify all service accounts cannot login into windows locally:
- 8.6.2 Log into the HMI trying to use any of the service accounts.
- 8.6.3 A message stating the Account cannot be log in should appear.

Description	Comments & Observations
Verify these accounts cannot be used to log into Windows: • “Deny Log on Locally” is set	
Expected Result to pass:	
Login should fail with a message stating the account is a service Account. Record the name of the account(s) tested.	

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8.7 Verify Service Account Password Policy

This task is to verify all service accounts have their password set to never expire:

- 8.7.1 Log into any of the DC using an administrator account. Open Server Manager, on the top menu chose "Tools-> Active Directory Users and Computers".
- 8.7.2 Domain Name (HMI.local)->GEV Controls->GEV Users and Computers" Select "Tools->Active Directory Users and Computers" and identify a service Account.
- 8.7.3 Double click it and under" Account" tab (in the Account Options), make sure "Password Never Expires" is checked.

Description	Comments & Observations
Verify these accounts have their passwords set to never expire.	
Expected Result to pass:	
Each Account should show the 'Password never expires' checked. Record the name of the account(s) tested.	

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8.8 Domain Password Policy Settings

This task is to verify the Domain Password Policies are setup correctly:

8.8.1 Log into any of the Domain controllers using admin accounts.

8.8.2 Open "Server Manager".

8.8.3 Select "Tools -> Group Policy Management".

8.8.4 Expand "Forest: Domain Name (HMI.local) -> Domains -> Domain Name (HMI.local)-> GEV Site Customization Policy

" And click "Edit". Expand the following path "Computer configuration-> Policies-> Windows Settings->Security Settings -> Account Policies -> Password Policy".

8.8.5 Verify each of the items listed below and make sure the settings are correct on the policies mentioned above:

Enforce Password History = 24 Passwords

Maximum Password Age = 90 Days

Minimum Password Age = 1 Day

Minimum Password Length = 8 Characters

Password must meet complexity requirements = Enabled

Description	Comments & Observations
Verify OTArmor Domain Policy password policy is set to: Enforce Password History = 24 Passwords Maximum Password Age = 90 Days Minimum Password Age = 1 Day Minimum Password Length = 8 Characters Password must meet complexity requirements = Enabled	
Expected Result to pass:	
Make sure each setting is correctly set as per list above.	

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8.9 Verify Password Policy on Domain Accounts

This task is to verify the Password Policy is enforcing the passwords in new domain accounts:

- 8.9.1 Log into any of the DC using an administrator account. Open Server Manager, on the top menu chose "Tools-> Active Directory Users and Computers".
- 8.9.2 A new window will appear (Active Directory Users and computers), expand "Domain Name (HMI.local)->GEV Controls->GEV Users and Computers" Right click on "GEV Users and Computers", then choose "New -> User".
- 8.9.3 Select a name for the username and click "Next". Enter a simple password to test the enforcement and click "Next" and then "Finish". An error pops up will appear stating the password does not meet the Password Policy requirement.
- 8.9.4 Attempt to set the user password again, this time apply a complex password meeting the settings on previous step.
- 8.9.5 Verify the user password is accepted.
- 8.9.6 Delete the account after the password is verified.

Description	Comments & Observations
Create a dummy domain account and test password policy is being enforced. Delete account when finished.	
Expected Result to pass:	
Verify the account can be created when the password meets the password enforcement established on the Domain.	

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8.10 Verify Password Policy on Local Accounts

This task is to verify Password Policy enforcing the password in new local accounts:

- 8.10.1 Log into an HMI using admin account.
- 8.10.2 Open “Control Panel -> All Control Panel Items-> Administrative Tools -> Computer Management -> Local Users and Groups -> Users”. Right click on “Users”, then choose “New User”.
- 8.10.3 Select a name for the username and click “Next”. Enter a simple password to test the enforcement and click “Next” and then “Finish”. An error pops up will appear stating the password does not meet the Password Policy requirement.
- 8.10.4 Attempt to set the user password again, this time apply a complex password meeting the Password Policy settings above.
- 8.10.5 Verify the user password is accepted.
- 8.10.6 Delete the account after the password is verified.

Description	Comments & Observations
Create a dummy HMI local account and test password policy is being enforced. Delete account when finished.	
Expected Result to pass:	
Verify the account can be created when the password meets the password enforcement criteria established.	

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8.11 Account Lockout

This task is to verify any user can be locked out after 10 failure login attempts:

- 8.11.1 Choose an account other than the Administrator account and attempt to login to a server with a wrong password. Repeat this step 10 times.
- 8.11.2 After the 10th failure, attempt to login with the correct password. A message should appear stating the Account has been locked due to many unsuccessful attempts.
- 8.11.3 Log into any of the DC using an administrator account. Open Server Manager, on the top menu chose "Tools-> Active Directory Users and Computers.
- 8.11.4 A new window will appear (Active Directory Users and computers), expand "Domain Name (HMI.local)->GEV Controls->GEV Users and Computers" Look for the User used on first step.
- 8.11.5 Once you find the User, right click on it and choose "Properties". Under the account tab, a message will appear stating the account is locked.
- 8.11.6 Log out of the Domain Controller.

Description	Comments & Observations
Verify if the incorrect password is entered more than 10 times a user account is locked out.	Account Tested: _____
Expected Result to pass:	
Verify the account is locked out after 10 unsuccessful attempts.	

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8.12 Verify Account Reset after Lockout

This task is to verify any user locked out can be reset and log in again:

8.12.1 Identify the account locked out in Test cases section 9.12

8.12.2 Log into any of the DC using an administrator account. Open Server Manager, on the top menu chose "Tools-> Active Directory Users and Computers".

8.12.3 A new window will appear (Active Directory Users and computers), expand "Domain Name (HMI.local)->GEV Controls->GEV Users and Computers". Look for the User you use on first step.

8.12.4 Once you find the User, right click on it and choose "Properties". Under the account tab, a message will appear stating the account is locked

8.12.5 Check the "Unlock Account" and click OK.

8.12.6 Try to login with this user with the correct password. This time you should be able to login.

Description	Comments & Observations
Verify account can be reset after lockout.	Account Tested: _____
Expected Result to pass:	
Verify the account is reset and can login again.	

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9.0 Windows Patching

9.1 Patch Scan and Deployment

This task is to verify the latest PVP disk has been deployed in PM1:

- 9.1.1 If you have application control, make sure to change Policy to Patching or Update mode in Trellix EPO. If not, directly proceed with the following steps.
- 9.1.2 Obtain the updated PVP patch disk or associated .iso file.
- 9.1.3 Mount the patch disk and execute the update batch file.
- 9.1.4 Verify the process completes successfully.

Description	Comments & Observations
Verify latest GE approved patches are deployed to the Ivanti Console. Update discs will always be one month behind.	Version Used: _____
Expected Result to pass:	
Verify the PVP disk run successfully.	

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9.2 Verify Device Detection for Patching

This task is to verify all the devices that are detected by Ivanti in PM1 Server:

9.2.1 Log into the PM1 server using an administrator account and launch the Ivanti shortcut on the desktop.

9.2.2 Perform a patch scan of the Domain using the pre-defined GE patch list.

9.2.3 Verify Ivanti lists all the HMIs, Servers (Virtual Machines) in the system.

9.2.4 Machine Groups as follows

11.3.4.1 OTArmor Servers Group:

- CA1
- DC1
- DR1
- PM1
- SP1

11.3.4.2 OTArmor Early Adopters:

- DC2

11.3.4.3 HMIs:

- All other physical and virtual machines

Description	Comments & Observations
Verify all HMIs, servers: virtual machines are reporting in the Ivanti console.	
Expected Result to pass:	
After the scan, verify all the devices are listed.	

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9.3 Verify Patch Deployment

This task is to verify all the devices that are detected by Ivanti in PM1 Server, and we can deploy patches into them:

- 9.3.1 Log into the PM1 server using an administrator account and launch the Ivanti shortcut on the desktop.
- 9.3.2 Perform a patch scan of the Domain using the pre-defined GE patch list.
- 9.3.3 Verify it lists all the HMIs, Servers (Virtual Machines) in the system.
- 9.3.4 Deploy all missing patches in a chosen HMI/Server. Verify that a "Patching in progress" message is displayed on the target machines.
- 9.3.5 If necessary, reboot the systems being patched when the patches complete deploying.
- 9.3.6 Perform a patch scan of the HMI/Server chosen using the pre-defined GE patch list, verify that the patches deployed no longer show as required.
- 9.3.7 If any new patches show as required, deploy the patches to the systems. If necessary, reboot the systems being patched when the patches complete deploying.
- 9.3.8 Repeat this process and verify the systems can be patched with no missing patches present.

Description	Comments & Observations
Scan the network and deploy any missing patches or service packs. Reboot HMIs/Servers as necessary.	
Expected Result to pass:	
Verify patches can be deployed in devices listed on Ivanti console.	

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10.0 HMI Backup & Restore

10.1 Perform Backups

This task is to verify Acronis has the backup plans set for all HMIs\Servers:

10.1.1 Log into the DR1 system using an administrator account and connect to Acronis Management Server.

10.1.2 Select Backup Plans.

10.1.3 Make sure we have defined a Backup Plan for OTArmor Servers.

10.1.4 Make sure we have defined a Backup Plan for HMIs

10.1.5 On each backup plan verify the configuration of it. Right click on the backup plan and choose "Edit".

10.1.6 The backup plan configuration determines the rate of backups, as well as how long they are kept. **There are three types of backups:**

12.1.5.1 **Full** – A complete backup of the machine.

12.1.5.2 **Differential** – A backup of changes that have occurred since the last Full backup.

12.1.5.3 **Incremental** – A backup of changes that have occurred since the last Differential backup.

10.1.7 There is a plan present for every asset.

10.1.8 All plans present a green status.

10.1.9 What to back up: Entire Machine

10.1.10 Where to back up: //NAS/Backup

10.1.11 How long to keep:

10.1.11.1 Full: 6 Months

10.1.11.2 Differential: 4 Weeks

10.1.11.3 Incremental: 7 Days

Description	Comments & Observations
Verify all HMIs/Servers are in the backup plan in the Acronis Management Console.	
Expected Result to pass:	
Verify back up plans were created that includes all the Windows devices.	

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10.2 Verify Restore Activities for Files/Folders. (Optional)

This task is to verify Acronis can restore Files /Folders.

10.2.1 Log into the DR1 system using an administrator account and connect to the Acronis Management Server.

10.2.2 Select DR1 on the All-Devices page.

10.2.3 Choose the Recovery > Recover... > Files/Folders option on the right:

10.2.4 Select C:\bginfo\ and select Download from the right.

10.2.5 Once the download is completed, verify that the folder can be opened, and the image inside can be viewed.

Description	Comments & Observations
Verify an HMI files/Folders can be restored in case of a disaster recovery by using the Acronis backup	HMI Restored: _____
Expected Result to pass:	
Verify Acronis Restore executes successfully. Take note of the Files/Folders restored under Comments & Observations.	

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10.3 Verify System After Restore.

This task is to verify functionality after the restore:

10.3.1 Log into the HMI/Server restored using an administrator account.

10.3.2 Verify the name of the HMI and IP Address.

10.3.3 Log into PM1 using an administrator account, verify Ivanti can scan the HMI/Server..

10.3.4 Log into LM1 using an administrator account, verify Splunk can receive logs from the HMI/Server.

10.3.5 Log into DR1 using an administrator account, verify Acronis can scan the HMI/Server.

Description	Comments & Observations
Verify the following after the restore is complete: - Machine name and IP address are correct - HMI is reporting in McAfee/Trend Micro/Splunk, Acronis and Ivanti.	
Expected Result to pass:	
Verify name and IP Address are correct in the HMI restored. Verify HMI is recognized by our different Server applications.	

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11.0 NETWORK SECURITY MONITORING (Optional)

11.1 Monitoring Port

11.1.1 Physically verify Nozomi device connections.

11.1.2 Verify if the ports are connected to a span port in the switches.

11.1.3 Connect to the switches through DR1/CatTools and verify if that port is a SPAN Port.

Description	Comments & Observations
Ensure that Nozomi Guardian ports are connected to the span port of network switch.	
Expected Result to pass:	
Verify SPAN port is configured in switch through CatTools and is there's a connection to the Nozomi Guardian as per Topology.	

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11.2 Nozomi Dashboard

11.2.1 Open a web browser and connect to Nozomi.

11.2.2 Verify if dashboard is populating data.

Description	Comments & Observations
Verify if data is populated in the Dashboard (Assets, Nodes, Sessions).	
Expected Result to pass:	
Data is shown in Nozomi Dashboard.	

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11.3 Network Traffic Validation

This task is to Verify the appliance is observing Network Traffic

11.3.1 On the web GUI, go to Administration > Network Interfaces. Make sure that the Throughput graph is being updated live with network data from all the applicable monitoring interfaces.

11.3.2 On the web GUI, go to Environment > Network view > Traffic tab. Make sure that the graphs are getting populated with live data information.

Description	Comments & Observations
Verify if the network traffic in network view Traffic tab populated live data information	
Expected Result to pass:	
Can see live data information in traffic tab.	

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11.4 Passive Monitoring and Network modeling and visualization Operation

This task is to verify that Nozomi IDS appliances show assets in the network with correct information.

11.4.1 On the Network View > Nodes tab, verify that this data is getting populated:

- Address
- Role
- Type
- MAC address
- Protocols
- On the Asset view, verify that this data is getting populated:
- Name
- Address
- Role
- Type
- OS/Firmware

11.4.2 On the Network View > Links tab, verify this data is getting populated:

- From
- To
- Protocol
- First Activity
- Last Activity
- Throughput

11.4.3 On the Network View > Session's tab, verify that this data is getting populated:

- Status (review that you have sessions with Status SYN, ACTIVE and CLOSED)
- From
- To
- Protocol

11.4.4 On the Network View > Graph tab, check that the network graph is being displayed. Apply various filters

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Description	Comments & Observations
Verify that Nozomi IDS appliance shows assets in the network with correct information	
Expected Result to pass:	
Can see live data populated in network view	

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11.5 Learning Mode Configuration

This task is to verify Nozomi IDS appliance is configured in Learning mode.

11.5.1 Login to Nozomi web interface

11.5.2 Select Administration-> Security Control Panel->

11.5.3 Click on “1-Learning” on left-hand pane

11.5.4 Under “Phase Switching”, select “Two Phase” and then “Learning”

11.5.5 Click “Save”

Description	Comments & Observations
Verify that Nozomi IDS appliance is configured in Learning mode	
Expected Result to pass:	
Can see Nozomi IDS appliance is configured learning mode	

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11.6 Malicious Signature detection

11.6.1 Open a web browser and connect to Nozomi.

11.6.2 Copy Antivirus file and proceed to copy through the network to another asset.

11.6.3 Verify Nozomi triggers an alert.

Description	Comments & Observations
Verify the popup warning is presented after copying the Test Virus file to a remote system.	
Expected Result to pass:	
Alert triggered after copy of virus file via network.	

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12.0 TRELLIX

12.1 Trellix Agent

This task is to verify all the devices that are detected by Trellix ePO in SP1:

12.1.1 Log into the SP1 using an administrator account and launch the Trellix ePO web management interface shortcut on the desktop.

12.1.2 Log in using the ePO administrator account.

12.1.3 Confirm that the machines with the System Tree are organized into the correct groups and appear as ePO managed endpoints

12.1.3.1 HMIs

12.1.3.2 OTArmor Servers

Description	Comments & Observations
Verify all HMIs and Virtual machines are reporting in the Trellix Management Console.	
Expected Result to pass:	
Verify all the devices are listed in the Trellix ePO management console.	

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12.2 Verify ePO Virus Definitions

This task is to update the virus definitions and verify the version of AV in Trellix ePO in SP1

- 12.2.1 Log into the SP1 using an administrator account and launch the Trellix ePO web management interface shortcut on the desktop.
- 12.2.2 Log in using the ePO administrator account.
- 12.2.3 Go to “Menu -> Software-> Master Repository”. Click “Check In Package”.
- 12.2.4 Select Product or Update (.zip) as the package type and browse to select the package with the new definitions and select “Open”
- 12.2.5 Click Next to view the package check-in summary information.
- 12.2.6 Select “Branch -> Current”.
- 12.2.7 Click “Save” to begin the check-in. Wait until the Check-in is complete and click Close.
- 12.2.8 Ensure the new virus definition package is in the repository:
 - 12.2.8.1 Click “Menu->Software->Master Repository”.
 - 12.2.8.2 Search the DAT in the name column.
 - 12.2.8.3 Verify that the DAT version displayed is the same as the version you checked in.

Description	Comments & Observations
Verify the latest AV signatures are installed into the management console.	Version Used: _____
Expected Result to pass:	
Verify the version of AV in Trellix console.	

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12.3 Verify ePO Software Deployment

This task is to verify the versions of AV on each HMI/Server match the Trellix ePO in SP1:

- 12.3.1 Log into the SP1 using an administrator account and launch the Trellix ePO web management interface shortcut on the desktop.
- 12.3.2 Log in using the ePO administrator account.
- 12.3.3 Go to "System Tree View -> Systems tab-> Preset: This Group and all Subgroups". Click on the name of the HMI/Server to verify and choose the "Products" tab.
- 12.3.4 A list of the products and the version of each will be listed.
- 12.3.5 Now login into the HMI/Server. In the bottom right in the task bar, a Trellix icon should be present.
- 12.3.6 Right click on the Trellix icon and click "About". A list of all the products installed will appear. Verify the versions match the ePO console Products tab for Trellix Data Exchange Layer, Trellix Application Control, Trellix Agent and Trellix DLP Endpoint.
- 12.3.7 Right click on the Trellix icon and click "Trellix Endpoint Security -> About". A list of all the products installed will appear. Verify the versions match the ePO console -> Products tab for Endpoint Security Platform and Threat Prevention.

Description	Comments & Observations
Verify all machines are up to date with management console. Check locally each HMI has the same definition level as management console.	
Expected Result to pass:	
Verify the version of AV matches from HMI/Servers to the ePO console.	

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12.4 Verify Antivirus Software Operation

This task is to verify viruses that are being detected by the HMI/Server and Trellix ePO in SP1 and they can be removed from the system:

12.4.1 Log into an HMI/Server using an administrator account. Open file explorer and type this path: [\\ca1\Admin Procedures\OTArmor](#). Locate the “Test-Antivirus” and copy it into the desktop of the HMI/Server to be tested.

12.4.2 Right click on “Test-Antivirus” and choose “Run with Powershell”. Once the Powershell was run a pop up should appear from Trellix stating a virus was detected.

Note: If the system has Solidcore installed, it will be necessary to add the powershell script to Solidcore rules to be allowed to run in “Enabled” mode.

12.4.3 Log into the SP1 server using an administrator account and launch the ePO web management interface shortcut on the desktop.

12.4.4 Log in using the ePO administrator account.

12.4.5 Go to “Tree View -> System tab-> Preset: This Group and all Subgroups”. Click on the name of the HMI/Server to verify and choose the “Threat Events” tab.

12.4.6 A list of threat will be presented, select the one generated as part of the testing and click on “Actions -> Delete”.

Description	Comments & Observations
Apply a test virus to an HMI. Verify is detected on the HMI, in the Trellix Management Console.	
Expected Result to pass:	
Verify the virus is detected by Trellix in the HMI/Server and ePO.	

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12.5 Verify Threat Event on LM1 (optional)

12.5.1 Log into the LM1 server using an administrator account and launch the Splunk web management interface shortcut on the desktop.

12.5.2 Select “OTArmor Visability” in the menu.

12.5.3 Select “Endpoint Protection”.

12.5.4 Verify that the “Antivirus Events” counter increments.

12.5.5 Click on the “Antivirus Events” number and verify the “Antivirus Event Detail” displays the threat event that was generated in the previous section.

Description	Comments & Observations
Apply a test virus to an HMI. Verify the threat event is displayed in the “Nexus OTArmor – Visibility” console.	
Expected Result to pass:	
Verify the threat event is detected is displayed in the Visibility console.	

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12.6 Verify Device Control Policy

This task is to verify a policy that exists to control USB Devices in Trellix ePO in SP1:

- 12.6.1 Log into the SP1 using an administrator account and launch the ePO web management interface shortcut on the desktop.
- 12.6.2 Log in using the ePO administrator account.
- 12.6.3 Go to “Menu -> Policy Catalog ->Data Loss Prevention x.x ”. Verify the DLP Policy “GEV Block All Devices” exists.

Description	Comments & Observations
Verify there is a device control policy that closes USB ports.	
Expected Result to pass:	
Verify the policy exists in ePO Applied to HMI's & OTArmor server which are subgroups under system tree.	

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12.7 Verify USB Device Access

This task is to verify you can access USB devices in HMI/Server with the correct user account:

12.7.1 Log into an HMI/Server using the “admin” account. Insert any encrypted USB to be tested.

12.7.2 Verify under this account you can access the USB and its contents.

Description	Comments & Observations
Verify USB devices can be accessed when an HMI/Server logs in with the exception user account.	HMI Tested: _____
Expected Result to pass:	
Verify access to USB devices is available for the exception user Account.	

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12.8 Verify USB Device Restrictions

This task is to verify access to USB devices is restricted for users without authorization:

- 12.8.1 Log into an HMI using the “GEMaint” account. Insert an encrypted USB to be tested.
- 12.8.2 Under this account, a pop up should appear explaining DLP has blocked the device.
- 12.8.3 Attempt to access files on the device and verify an “Access Denied” message is received when trying to get the USB’s contents.
- 12.8.4 Log into the SP1 server using an administrator account and launch the ePO web management interface shortcut on the desktop.
- 12.8.5 Log in using the ePO administrator account.
- 12.8.6 Go to Menu -> Policy Catalog -> Data Loss Prevention 11.X -> under policy - GEV Block All Devices”. Select “Edit -> GEV Removable Storage Device Rule -> Device Control -> GEV Block Removable Device Rule, choose “Exceptions” tab and choose “Excluded Users”. Verify that the GEMaint account does not have a USB exception.

Description	Comments & Observations
Verify USB devices can NOT be accessed when an HMI logs in with a different user account from the exception.	
Expected Result to pass:	
Verify there’s NOT access to USB devices with the incorrect exception User Account.	

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12.9 Verify USB Device Restrictions Event on LM1 (Optional)

- 12.9.1 Log into the SP1 server using an administrator account and launch the ePO web management interface shortcut on the desktop.
- 12.9.2 Log in using the ePO administrator account.
- 12.9.3 Go to "Menu -> Automation -> Server Tasks".
- 12.9.4 Select "Run" on the "DLP Block Events" task.
- 12.9.5 Verify the "DLP Block Events" task status is "Completed".
- 12.9.6 Log into the LM1 server using an administrator account and launch the Splunk web management interface shortcut on the desktop.
- 12.9.7 Select "OTArmor Visibility" in the menu.
- 12.9.8 Select "Endpoint Protection".
- 12.9.9 Verify that the "DLP" counter increments.
- 12.9.10 Click on the "DLP Events" number and verify the "DLP Event Detail" displays the USB block event that was generated in the previous section.

Description	Comments & Observations
Verify the USB Device Restriction event is displayed in the "Nexus OTArmor – Visibility" console.	
Expected Result to pass:	
Verify the USB Device Restriction is detected displayed in the Visibility console.	

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12.10 SolidCore

This task is to verify SolidCore is working properly.

12.10.1 Changing Status

- 12.10.1.1 Log into each HMI/Server with an administrator account.
- 12.10.1.2 Open an administrator command prompt and verify the system is solidified, in enable mode, in lock down mode and is managed by the ePO server
- 12.10.1.3 Log into the SP1 with administrator credentials and launch the ePO web management interface shortcut on the desktop.
- 12.10.1.4 Log in using the ePO administrator account.
- 12.10.1.5 Go to "System Tree", click on "My Organization" and select all the devices listed and under "Actions" select "Agent"-> "Run Client Task Now". A new screen will show up, on the left pane select "SolidCore 8.X.X". At the middle pane select "Begin Update Mode". At the right pane select "Create new task" and click on "Run Task Now".
- 12.10.1.6 Wait until the task completes.
- 12.10.1.7 On HMI/Server, open an administrator command prompt and verify the system is in Update mode.
- 12.10.1.8 Repeat steps to change into different modes

12.10.2 Switching between modes

- 12.10.2.1 From Observe mode, you can switch to Enabled or Disabled mode.
- 12.10.2.2 From Enabled mode, you can switch to Disabled, Update, or Observe mode.
- 12.10.2.3 From Update mode, you can switch to Enabled or Disabled mode.
- 12.10.2.4 From Disabled mode, you can switch to Enabled, Update, or Observe mode.

Description	Comments & Observations
Verify the change on the mode status is working.	
Expected Result to pass:	
Verify change of status executes successfully.	

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12.11 Testing SolidCore Application

- 12.11.1.1 Log into the DR1 Server with an administrator account.
- 12.11.1.2 Open an administrator command prompt and verify the system is solidified, in enable mode, in lock down mode, and is managed by the ePO server.
- 12.11.1.3 Locate the “putty.exe” file and copy the application to a HMI’s desktop.
- 12.11.1.4 On the HMI, open an administrator command prompt and verify the HMI is solidified, in enable mode, in lock down mode, and is managed by the ePO server.
- 12.11.1.5 Attempt to execute the putty application on the desktop. Verify that the application is blocked by whitelisting.
- 12.11.1.6 From the ePO server, place the HMI in update mode.
- 12.11.1.7 Verify the HMI is in update mode. Erase and copy Putty again and attempt to run the application on the desktop. Verify the file is permitted to be executed.
- 12.11.1.8 From the ePO server, remove the system from update mode to enable mode. Verify the HMI that it is in enable mode.
- 12.11.1.9 Attempt to execute the Putty application on the desktop, verify that the application is permitted to execute.

Description	Comments & Observations
Verify Putty cannot be run unless in update mode.	
Expected Result to pass:	
Verify running Putty gets denied by solidcore and allows it when in update mode.	

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13.0 Controller Hardening

13.1 Secure Mode

This task is to verify the Mark VIe controllers can be put into Secure Mode.

13.1.1 In Toolbox, verify the Mark VIe controllers can be put into Secure Mode.

13.1.2 In CA1, verify the CA1 server is issuing a certificate for the controllers.

14.1.3 Remove controller from secure mode.

Description	Comments & Observations
Verify the Mark VIe Controller can be put into Secure Mode. Ensure CA1 is issuing certificates for the controller.	
Expected Result to pass:	
Verify that the Mark VIe Controller can be put into Secure Mode, CA1 is issuing certificates for the controller.	

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14.0 **SPLUNK** (Optional)

14.1 Verify Devices are reporting to Splunk

- 14.1.1 Log into LM1 server using an administrator account and launch the Splunk web management interface shortcut on the desktop.
- 14.1.2 Click "Search" and then click on "Data Summary".
- 14.1.3 A window will pop up with all the devices sending logs to Splunk:
Servers, HMI's, Switches, Controllers, Firewalls and Hypervisors.
- 14.1.4 Verify all the devices listed are expected and there's no devices missing. (Devices present and connected in the network). Also make sure we receive logs for each device and the time is current.

Description	Comments & Observations
Verify all devices are reporting to Splunk. This includes all HMI's, Virtual machines, switches, and firewall.	
Expected Result to pass:	
Verify Splunk is receiving "logs" from the devices connected on the network successfully.	

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14.2 Verify Successful/Unsuccessful login reporting

- 14.2.1 If not generated previously, generate several successful and failed logins on different devices logging to the Splunk.
- 14.2.2 Log into LM1 server using an administrator account and launch the Splunk web management interface shortcut on the desktop.
- 14.2.3 From the Splunk console -> select the GEV OTArmor™.
- 14.2.4 Verify the Controller panels are populated. The Windows Active Directory panel will alert on the following Windows event codes:
- Event code 4625: Login Failed
 - Event code 4624: Login Success

Description	Comments & Observations
Verify successful and unsuccessful login attempts are captured in the Splunk as a troubleshooting tool.	
Expected Result to pass:	
Verify Splunk is receiving “logon failure” and “logon success” from the devices connected on the network.	

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14.3 Verify NIDS is reporting

- 14.3.1 If not generated previously, generate several NIDS strings in the system (Test-NIDS.ps1) on any of the HMIs and take note of the time they were executed. This powershell will simulate an attack in the system.
- 14.3.2 Log into the firewall and under Network->Interfaces check the port is configured for NIDS.
- 14.3.3 Also, under Log&Report->Intrusion Prevention, verify you got the logs for the attack (Test-NIDS.ps1).
- 14.3.4 Log into LM1 server using an administrator account and launch the Splunk web management interface shortcut on the desktop.
- 14.3.5 Click "Search & Reporting" and then on "Search" type "GEV NIDS TEST STRING".
- 14.3.6 A list of logs will appear with the string on it. Make sure the times match with the ones you executed and from the devices they were executed.

Description	Comments & Observations
Verify "GEV NIDS TEST STRING" can be found in the logs after executing Test-NIDS.ps1.	
Expected Result to pass:	
Verify Splunk is receiving "GEV NIDS TEST STRING" logs and they match the ones that were executed with Test-NIDS.ps1.	

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15.0 HIGH AVAILABILITY FAIL-OVER

15.1.1 Verify VH1, VH2 and VH3 are running. All VM's on each hypervisor must be on and operating normally (Skyline Health good and make sure unhealth items in VSAN health).

15.1.2 Remove the HA Network connections from **one** of the Hypervisors (VH1 or VH2 or VH3).

15.1.3 Login into VSphere with admin credentials. Verify all the Virtual Machines associated with the Hypervisor chosen are migrating to their corresponding backup Hypervisors.

Note. This step takes time, the change is not instant.

15.1.4 Reconnect the Network connections on Hypervisor chosen on step 20.1.2.

15.1.5 Manually migrate all the Virtual machines that belong the chosen Hypervisor:

15.1.5.1 Right click on the VM, select "Edit Settings" and Uncheck the CD Drive. Click "Save".

15.1.5.2 Right click on the VM, select "Migrate" -> "Computer Resources" -> "Next"-> "Finish".

15.1.6 Verify the migration of the VM's involved have been successful.

Description	Comments & Observations
Verify the Virtual Machines associated with VHx move to its corresponding backup server. Return the network connections to VHx and manually migrate the VM's to their original Virtual Host. Verify the migration is executed successfully.	
Expected Result to pass:	
Verify HA fail-over is performed successfully. All VMs migrate to the rest of the Hypervisors. Verify after re-establishing, that all VM's can be migrated to their original Hypervisor.	

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16.0 AGREED RESIDUAL ACTIONS

Description:

Identify residual actions, questions, agreements, or any other issues to be addressed after the CWT but before installation starts. Include who will be doing the work and the estimated completion date. Additional residual action lists may be attached at the end of this document as needed.

Item #	Description of Change or Action	Action Item Owner	Estimated Completion Date
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			

GEV Approval	Date: _____	Initials _____
Customer Approval	Date: _____	Initials _____

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17.0 CUSTOMER WITNESS TEST SIGNATURE SHEET

The undersigned agree that the factory acceptance test was completed on the date indicated.

GEV <i>Approval</i>	CUSTOMER <i>Approval</i>
----- (signature)	----- (signature)
----- (date)	----- (date)

GEV CONTROL SOLUTIONS
For
SAUDI ELECTRICITY COMPANY CENTRAL REGION
RIYADH PP11
F6900G6
CUSTOMER WITNESS TEST

The following approvals constitute the satisfactory completion of the Customer Witness Test (CWT).
The CWT was conducted on a structured basis as outlined in the previous pages.
The above signatures constitute an agreement on the outstanding residual actions and a commitment by GEV to resolve and/or complete any outstanding action items. Outstanding items are identified on the Agreed Residual Action Items list or as an attachment to this document.
Customer's acceptance authorizes the billing associated with the Customer Witness Test completion.

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